AI-BASED EARLY MENTAL HEALTH BREAKDOWN DETECTION FROM SPEECH PATTERNS

MileStone-1 Report

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OBJECTIVE: BUILD A DUAL-MODAL SCREENING MODEL USING AUDIO + TRANSCRIPT FUSION.

CORE FOUNDATION MODELS

- WHISPER (OPENAI/WHISPER-SMALL / MEDIUM) – AUDIO ENCODER
- BIOCLINICAL-MODERNBERT (LINDVALLLAB) – LONG-CONTEXT TRANSCRIPT ENCODER
- ONLY THE FUSION LAYER AND CLASSIFIER ARE TRAINABLE

DESIGN STRATEGY

- KEEP LARGE MODELS FROZEN
- EXTRACT RICH EMBEDDINGS
- LEARN TASK-SPECIFIC CLASSIFIER HEAD
- IMPROVE GENERALIZATION AND REDUCE OVERFITTING

DUAL BRANCH PROCESSING

Audio Branch – Whisper

- Input: 15–20 sec audio segments (DAIC-WOZ)
- Extract final encoder hidden states
- Apply Temporal Statistics Pooling (mean + standard deviation)
- Captures prosody, pitch flattening, speech rate, pauses
- Whisper remains frozen

Transcript Branch – BioClinical-Modern-BERT

- Processes full transcripts (up to 8,192 tokens)
- Uses Flash Attention + RoPE
- [CLS] token used as text embedding
- Augmented with Linguistic Feature Vector:
  - Sentiment polarity
  - First-person pronoun density
  - Affective scores
  - Model remains frozen

FUSION MECHANISM

- Concatenate audio + text embeddings
- Project into shared 512-dimensional space
- Apply Sigmoid-based gating mechanism

ADAPTIVE BEHAVIOR

- Poor audio quality → prioritize text
- Short/laconic answers → prioritize acoustic features

CLASSIFIER HEAD

- 3 Fully Connected layers
- Dropout for regularization
- Weighted Cross-Entropy Loss
- Higher weight for “At-Risk” class to maximize screening sensitivity

SIGNAL EXTRACTION AND DATASETS

SIGNAL EXTRACTION LAYER

- Transforms embeddings into interpretable indicators

ACOUSTIC SIGNALS

- Reduced prosodic variability
- Extended pauses
- Low energy levels

LINGUISTIC SIGNALS

- Negative self-references
- PHQ-9 symptom mentions (sleep, appetite, fatigue)
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RETRIEVAL-AUGMENTED REASONING

- MiniLM retrieves APA/WHO guideline snippets
- Instruction-tuned LLM generates safe, non-diagnostic summary
- Recommends professional screening

DATASET INVENTORY

Synchronized Audio-Text pairs for Gated Fusion training.
~9,000+ vocal segments for high-performance scaling.

Dataset	Format	Size/Count	Primary Metric	Purpose
DAIC-WOZ	Audio/Text	189 Sessions	PHQ-8 Score	**Core Fusion Training**
MODMA	Audio/Text	52 Participants	Binary Diagnosis	**Clinical Alignment**
RAVDESS	Audio-Only	7,356 Files	8 Emotion Tiers	**Classifier Pre-training**
Kaggle	Audio-Only	1,200+ Files	Depressed/Normal	**Noise Robustness**
Custom Set	Audio/Text	~500 Samples	Stress Levels	**RAG/Reasoning Test**

HARDWARE & INFRASTRUCTURE
TITLE: COMPUTE ENVIRONMENT & RESOURCES

- Primary Training: Kaggle Notebooks (NVIDIA T4 – 16GB VRAM)
- Secondary Environment: Google Colab (Free Tier)
- ~30 GPU hours/week (Kaggle)
- 100GB persistent storage
- 12–15GB system RAM (Colab)

MODEL FEASIBILITY STRATEGY
TITLE: MODEL OPTIMIZATION APPROACH

Content:

- Core Model: Whisper + BioClinical-Modern-BERT
- Phase 1: Freeze backbone → Train classification head only
- Phase 2: Selective fine-tuning (if budget permits)
- Use:
- Mixed Precision (FP16)
- Gradient Checkpointing
- Small Batch Size (8–16)
- Gradient Accumulation

DATA PREPROCESSING PLAN
DATA HANDLING & STORAGE FEASIBILITY

- Content:
- 50+ hours DAIC-WOZ audio
- One-time offline feature extraction:
- Mel Spectrogram (STFT)
- Wav2Vec embeddings
- Features stored → No repeated computation
- Smaller datasets (RAVDESS, MODMA) → No constraints

EVALUATION FRAMEWORK OVERVIEW
TITLE: MULTI-DIMENSIONAL EVALUATION STRATEGY

We evaluate across:

- Classification Performance
- Severity Score Regression
- Explainability (XAI)
- Generalization & Robustness

CROSS-FUNCTIONAL EXPERTISE FOR MULTIMODAL FUSION

- Role 1: Multimodal Audio Engineer
  - Focus: Whisper Branch & Paralinguistic preservation.
  - Scope: Implementing whisper-medium encoder and Temporal Statistics Pooling for vocal jitter/pitch analysis.
- Role 2: Clinical NLP & LLM Architect
  - Focus: Transcript Branch & RAG System.
  - Scope: Managing Bio_Clinical-ModernBERT (8k context) and Retrieval-Augmented Reasoning via MiniLM.
- Role 3: Fusion & Optimization Lead
  - Focus: Sigmoid-Gated Integration & Class Imbalance.
  - Scope: Designing adaptive gating layers and Weighted Cross-Entropy Loss strategies to prevent modality collapse.
- Role 4: MLOps & Clinical Validation Lead
  - Focus: Pipeline Integration & XAI.
  - Scope: Monitoring F1-Score/Recall and developing the "Signal Extraction" interface for clinical indicators.

PROJECT ROADMAP

- Phase 1: Foundation (Feb 20 - Feb 26): SOTA review of Whisper/ModernBERT; establishing the F1 > 0.85 target benchmark.
- Phase 2: Synchronization (Feb 27 - Mar 5): Forced Alignment of DAIC-WOZ/MODMA audio-text segments (15–20s windows).
- Phase 3: Engineering (Mar 6 - Mar 19): Building dual-branch encoders and the Sigmoid-Gating 512D shared space.
- Phase 4: Optimization (Mar 20 - Mar 26): AdamW training loops and Modality Ablation tests to verify gate sensitivity.
- Phase 5: Validation (Mar 27 - Apr 2): LOSO Cross-Validation and error analysis on "False Neutrals"; RAG integration.
- Phase 6: Deployment (Apr 3 - Apr 16): Launching API and generating LLM-synthesized traceable clinical reports.

CONCLUSION

- Objective Diagnostics: Transitioning from subjective self-reporting to objective vocal biomarker detection.
- Foundation Fusion: Leveraging Whisper for acoustics and Bio_Clinical-ModernBERT for deep semantic symptom clusters.
- Explainable RAG: Moving beyond "Black Box" predictions to provide traceable summaries grounded in APA/WHO guidelines.
- The Vision: A non-invasive, scalable triage tool that identifies mental health distress signals before a clinical crisis occurs.

THANK YOU